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Department of Chemistry

BIOLOGY AND ENVIRONMENTAL SCIENCE (CY202)

UNIT-III : CELLS AND BIOMOLECULES

UNIT-2

1. Definition of Enzymes

- Enzymes are biological macromolecules (mostly proteins, some RNAs) that act as biocatalysts to accelerate biochemical reactions without being consumed in the process.

2. Characteristics of Enzymes

- Specificity: Highly specific for their substrates (lock-and-key or induced-fit models).
- Catalytic Efficiency: Accelerate reactions by lowering the activation energy.
- Reusable: Not consumed during the reaction; can be used repeatedly.
- Optimal Conditions: Function best under specific pH, temperature, and ionic strength.
- Regulation: Activity can be modulated by inhibitors, activators, or feedback mechanisms.

3. Mechanism of Action

1. Substrate binds to the enzyme's **active site**, forming an enzyme-substrate complex.
2. Enzyme stabilizes the transition state, reducing activation energy.
3. Products are released, and the enzyme is free to catalyze another reaction.

4. Types of Enzymes

- Based on the type of reaction catalyzed:
    1. **Oxidoreductases:** Catalyze oxidation-reduction reactions.
    2. **Transferases:** Transfer functional groups between molecules.
    3. **Hydrolases:** Catalyze hydrolysis reactions.
    4. **Lyases:** Break chemical bonds without water or oxidation.
    5. **Isomerases:** Rearrange atoms within a molecule.
    6. **Ligases:** Join two molecules using ATP.

5. Applications of Enzymes

- **Medical:** Diagnostics, drug production, and enzyme replacement therapy (e.g., lactase for lactose intolerance).
- **Industrial:** Used in food processing (amylase in bread making), textiles (cellulase for fabric softening), and biofuels (lipase in biodiesel production).
- **Research:** Tools in molecular biology (e.g., DNA polymerase in PCR).

6. Enzyme Inhibition

- **Competitive Inhibition:** Inhibitor competes with the substrate for the active site.
- **Non-Competitive Inhibition:** Inhibitor binds to a different site, altering enzyme activity.
- **Uncompetitive Inhibition:** Inhibitor binds only to the enzyme-substrate complex.

7. Importance of Enzymes

- Essential for life processes, including metabolism, DNA replication, and energy production.
- Highly efficient and environmentally friendly alternatives to chemical catalysts in industrial applications.

Reversible Enzyme Inhibition

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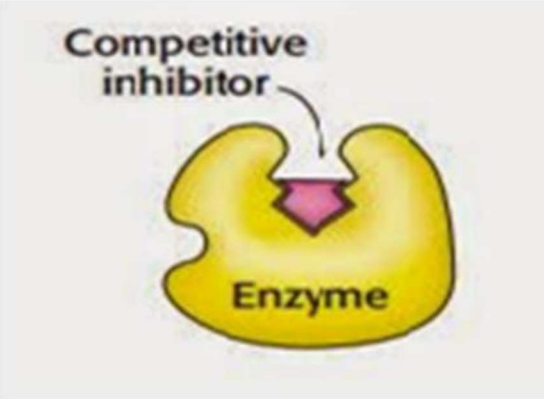
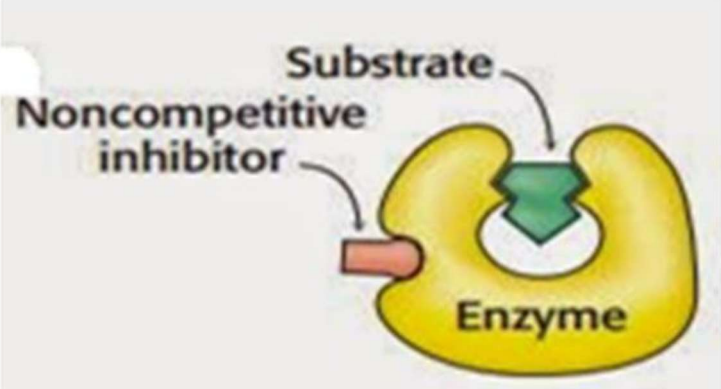
Competitive Inhibition



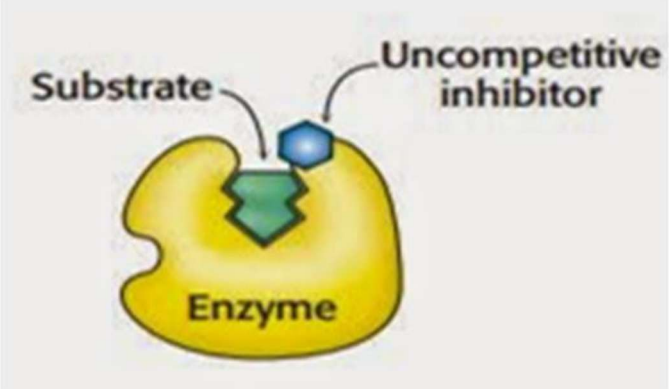
Inhibitor→Active site

Non-competitive inhibition



Inhibitor→Allosteric site

Uncompetitive inhibition



Inhibitor→ES complex

What Is Fermentation?

Fermentation is a biochemical process that gets energy from carbohydrates and does not require oxygen.

ethanol (wine)

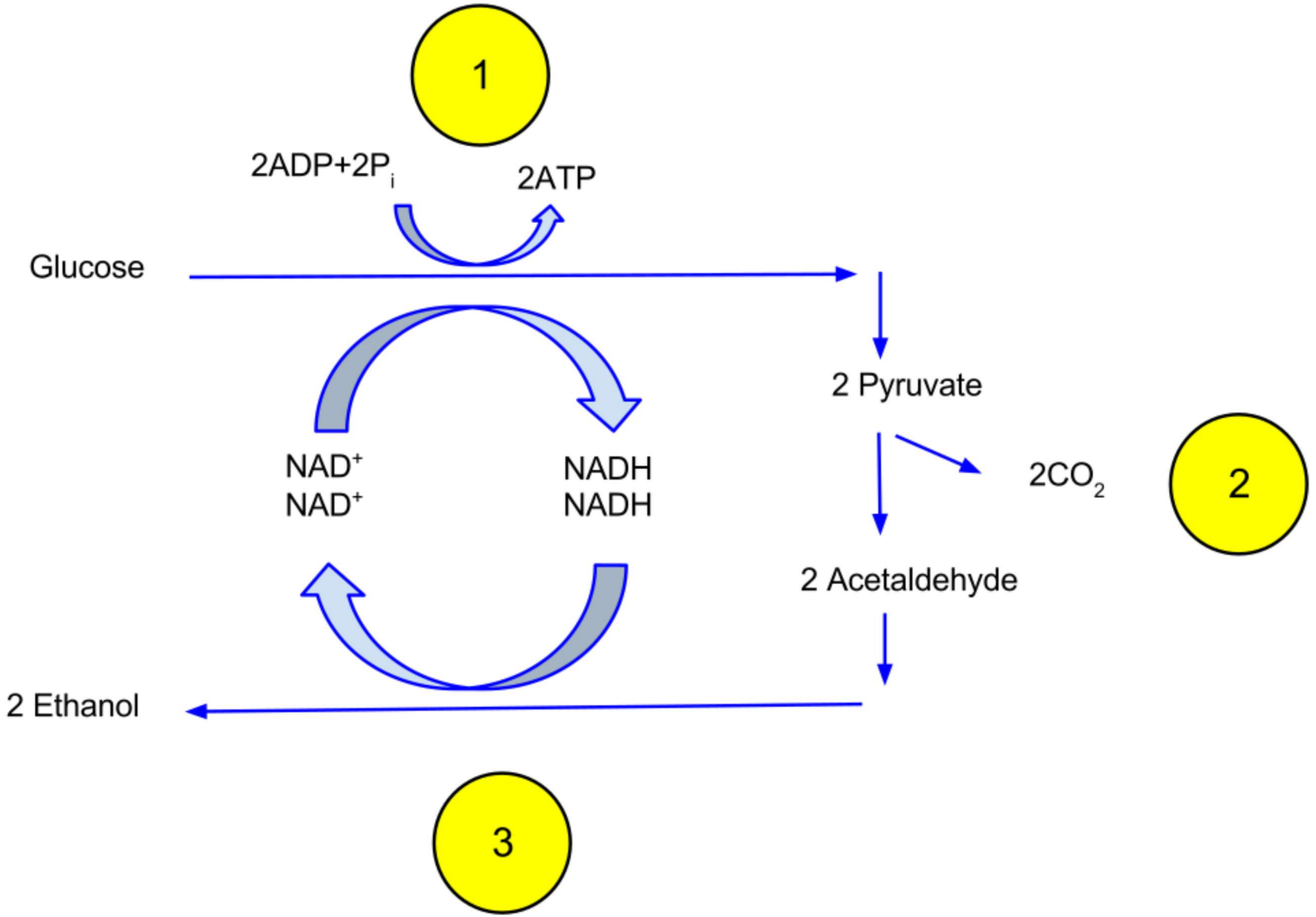
hydrogen (farts)

lactic acid (cheese)

carbon dioxide (bread)

glucose → ethanol + carbon dioxide + energy

$$C_6H_{12}O_6 \rightarrow 2\,C_2H_5OH + 2\,CO_2$$


1
2ADP+2P$_i$
2ATP
Glucose
2 Pyruvate
NAD$^+$
NAD$^+$
NADH
NADH
2CO$_2$
2
2 Acetaldehyde
2 Ethanol
3

Fermentation is a metabolic process in which microorganisms like bacteria, yeast, or molds convert sugars and other organic compounds into simpler molecules, often producing energy for the microorganisms in the process. It occurs in the absence of oxygen (anaerobic conditions) and is commonly used in food and beverage production, as well as in various industrial processes.

Here's a basic breakdown of how fermentation works:

1. Sugar Breakdown (Glycolysis)

Fermentation begins with the breakdown of glucose (or other sugars) in a process called **glycolysis**. In glycolysis, glucose is split into two molecules of pyruvate, producing a small amount of energy (ATP) and NADH, which is used to carry electrons.

2. Regeneration of NAD+

Since glycolysis requires NAD+ (nicotinamide adenine dinucleotide) to work, the cell needs to regenerate NAD+ to keep glycolysis going. In the absence of oxygen (anaerobic conditions), this regeneration happens through fermentation.

3. Fermentation Pathways

Depending on the organism, there are two main types of fermentation:

Alcoholic Fermentation (e.g., Yeast)

In yeast cells, pyruvate is converted into ethanol (alcohol) and carbon dioxide. The process also regenerates NAD+, which allows glycolysis to continue. This type of fermentation is used in the production of beer, wine, and bread.

•**Equation for Alcoholic Fermentation**:

Glucose → 2 Ethanol + 2 CO_2 + Energy (ATP)

•**Lactic Acid Fermentation (e.g., Muscle Cells, Some Bacteria)**

In muscle cells (during strenuous activity) and in certain bacteria, pyruvate is reduced to lactic acid (lactate). This also regenerates NAD+, allowing glycolysis to continue. Lactic acid fermentation is what occurs when muscles experience oxygen deprivation, leading to lactic acid buildup, which causes the "burn" feeling.

•**Equation for Lactic Acid Fermentation**:

Glucose → 2 Lactic Acid + Energy (ATP)

4. Byproducts

The byproducts of fermentation depend on the type of fermentation:

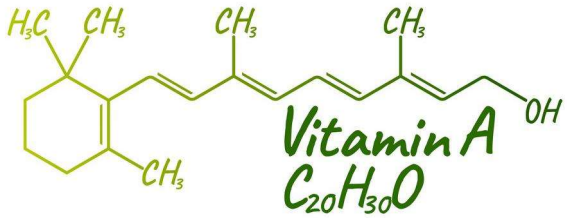
•**Alcoholic Fermentation**: Ethanol and CO_2 (important for brewing and baking).

•**Lactic Acid Fermentation**: Lactic acid (important for dairy products like yogurt and fermented foods).

5. Applications

Fermentation is widely used in food production, such as in the making of bread (where yeast ferments sugars to produce CO_2, causing the dough to rise), alcoholic beverages, and dairy products like yogurt and cheese. It also has industrial applications, including in the production of biofuels, antibiotics, and other chemicals.

Vitamin and mineral deficiencies can lead to a variety of diseases, each with distinct symptoms and effects on the body. Here's an overview of some common deficiencies and the diseases they cause:



1. Vitamin A Deficiency

•**Disease**: **Night Blindness** (and potentially complete blindness)

•**Symptoms**: Night blindness, dry eyes, damaged cornea, increased susceptibility to infections.

•**Cause**: Insufficient intake of vitamin A, which is crucial for maintaining healthy vision and immune function. Vitamin A is found in foods like carrots, liver, and leafy green vegetables.

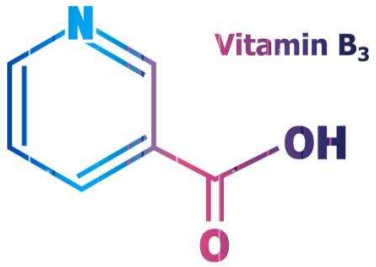
2. Vitamin B1 (Thiamine) Deficiency

•**Disease**: **Beriberi**

•**Symptoms**: Fatigue, weakness, nerve damage, edema (swelling), and heart problems. There are two forms of beriberi:

- Wet beriberi (affects the cardiovascular system)
- Dry beriberi (affects the nervous system)

•**Cause**: A lack of thiamine, which is essential for energy metabolism. Common in people with poor diet, particularly those who consume a lot of refined rice or alcohol. Found in whole grains, beans, and pork.

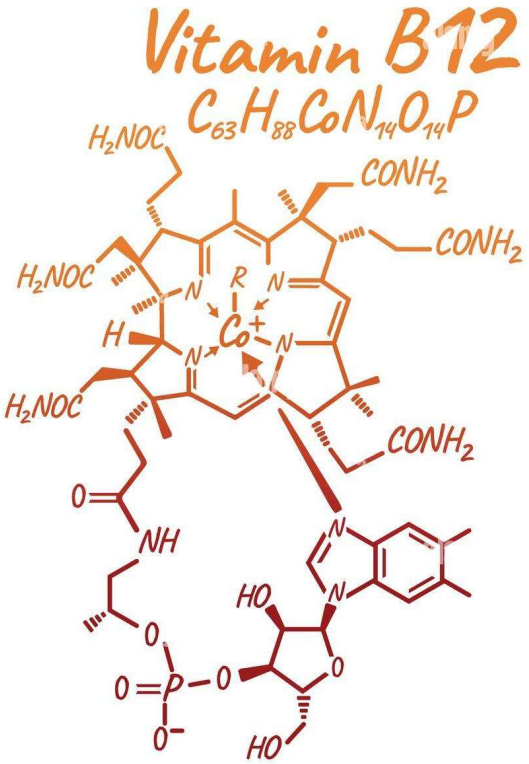


3. Vitamin B3 (Niacin) Deficiency

•**Disease**: **Pellagra**

•**Symptoms**: Dermatitis, diarrhea, dementia, and in severe cases, death (the "4 D's").

•**Cause**: Insufficient niacin, often associated with diets heavily reliant on corn and low in other niacin-rich foods like meat, fish, and legumes.



4. Vitamin B12 Deficiency

•**Disease**: **Pernicious Anemia**

•**Symptoms**: Fatigue, weakness, memory problems, tingling in the hands and feet, difficulty walking, and mood disturbances.

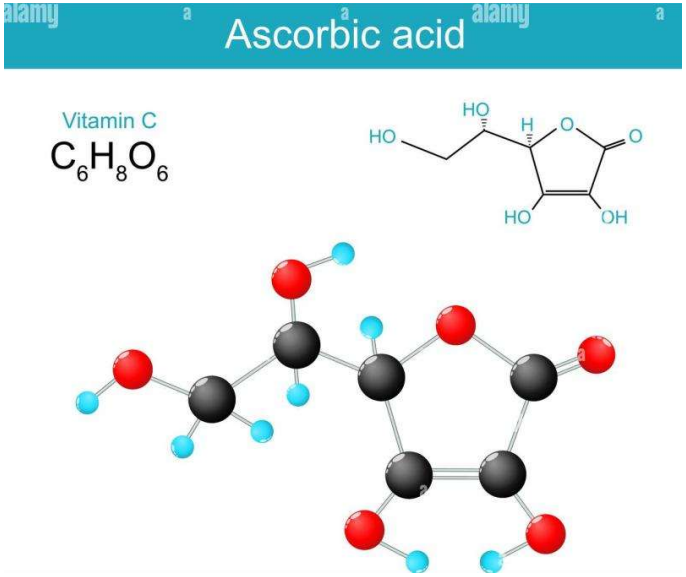
•**Cause**: Lack of vitamin B12, which is important for red blood cell formation and nerve function. Common in vegetarians and older adults, as B12 is primarily found in animal products.

5. Vitamin C Deficiency

Disease: **Scurvy**

Symptoms: Fatigue, swollen gums, joint pain, and skin problems (such as bruising). In severe cases, it can lead to bleeding and anemia.

Cause: Insufficient vitamin C, which is important for collagen synthesis and immune function. Found in citrus fruits, strawberries, and leafy greens.

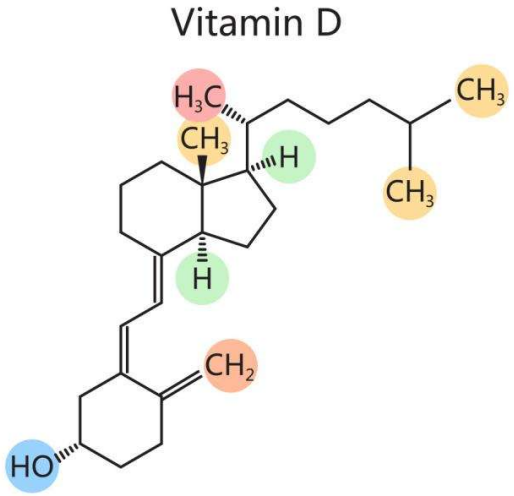


6. Vitamin D Deficiency

•**Disease**: **Rickets** (in children) / **Osteomalacia** (in adults)

•**Symptoms**: Bone deformities (in children), bone pain, muscle weakness, and fractures.

•**Cause**: Lack of vitamin D, which helps the body absorb calcium and promote bone health. It can be caused by insufficient sunlight exposure or inadequate dietary intake. Found in fortified foods, fatty fish, and eggs.

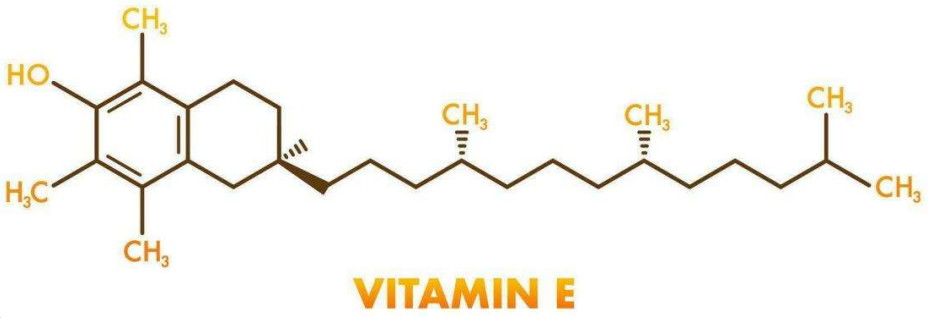


7. Vitamin E Deficiency

•**Disease**: **Neurological Problems**

•**Symptoms**: Muscle weakness, coordination problems, vision problems, and immune system issues.

•**Cause**: Lack of vitamin E, which acts as an antioxidant and protects cells from damage. Found in nuts, seeds, and vegetable oils.



8. Vitamin K Deficiency

Disease: **Bleeding Disorders**

Symptoms: Easy bruising, excessive bleeding from minor cuts, and excessive menstrual bleeding.

Cause: Insufficient vitamin K, which is vital for blood clotting. Found in leafy green vegetables, broccoli, and fermented foods.

9. Iron Deficiency

•**Disease**: **Iron-Deficiency Anemia**

•**Symptoms**: Fatigue, weakness, pale skin, dizziness, and headaches.

•**Cause**: Insufficient iron, which is necessary for the production of hemoglobin in red blood cells. Common in menstruating women, children, and those with poor dietary intake. Found in red meat, beans, and fortified cereals.

10. Calcium Deficiency

•**Disease**: **Osteoporosis** (in adults) / **Rickets** (in children)

•**Symptoms**: Brittle bones, muscle cramps, and in severe cases, bone fractures.

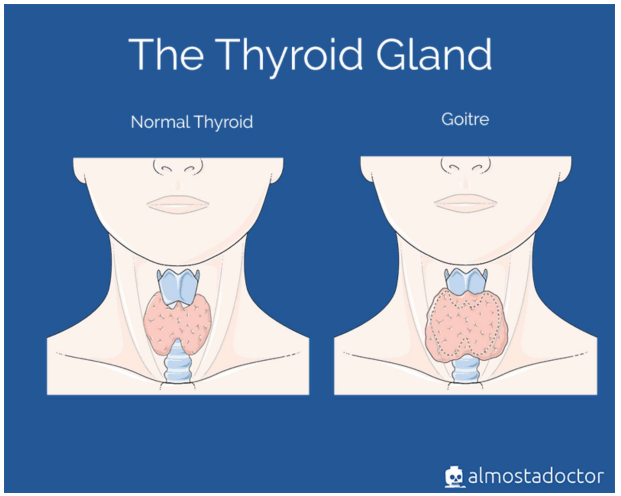
•**Cause**: Lack of calcium, which is essential for bone and teeth health. Found in dairy products, fortified plant milks, and leafy greens.

11. Iodine Deficiency

Disease: **Goiter**

Symptoms: Enlarged thyroid gland, fatigue, weight gain, and sensitivity to cold.

Cause: Lack of iodine, which is crucial for thyroid function. Iodine is found in iodized salt, seafood, and dairy products.



12. Folate (Vitamin B9) Deficiency

•**Disease**: **Megaloblastic Anemia**

•**Symptoms**: Fatigue, weakness, irritability, and neural tube defects in pregnant women (e.g., spina bifida).

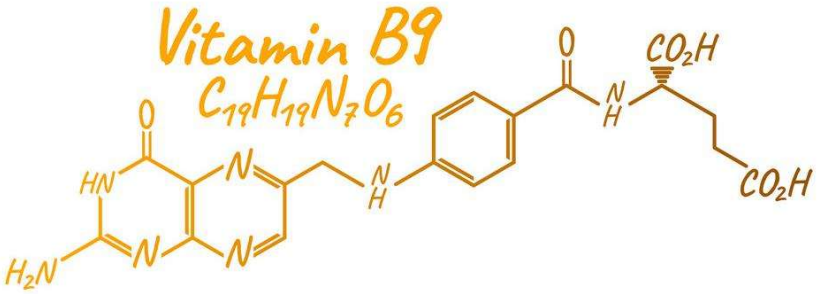
•**Cause**: Lack of folate, which is essential for DNA synthesis and red blood cell formation. Found in leafy vegetables, legumes, and fortified cereals.

13. Zinc Deficiency

•**Disease**: **Zinc Deficiency Disorders**

•**Symptoms**: Impaired immune function, hair loss, delayed wound healing, and skin rashes.

•**Cause**: Insufficient zinc, which is important for immune function and enzyme reactions. Found in meat, shellfish, legumes, and seeds.



Bioenergy is energy derived from biological sources, including plants, animals, and microorganisms. It plays a crucial role in renewable energy systems, as it can be used in various forms like heat, electricity, and fuel. Here's an outline of the key concepts related to bioenergy:

1. Definition of Bioenergy

Bioenergy refers to energy produced from organic materials, often referred to as **biomass**. This energy is harnessed from the chemical energy stored in plants, animals, and microorganisms. Unlike fossil fuels, bioenergy is renewable and can be replenished over time.

2. Sources of Bioenergy

Bioenergy can be derived from a wide range of sources, such as:

Wood and Forest Products: Wood, sawdust, and other forestry residues are common sources of bioenergy. They can be used directly for heating or transformed into biofuels.

Agricultural Residues: Crop residues like corn stalks, rice husks, and wheat straw are potential sources of bioenergy.

Dedicated Energy Crops: Certain crops like switchgrass, miscanthus, and algae are specifically grown for bioenergy production.

Animal Manure: Manure from livestock can be converted into biogas through anaerobic digestion, which produces methane.

Waste: Organic waste from households, industries, and landfills can be used to generate energy through various processes.

3. Forms of Bioenergy

Bioenergy can be converted into different forms of usable energy:

Biogas: Produced from the anaerobic decomposition of organic matter (such as animal waste and plant materials). Biogas primarily consists of methane (CH_4) and can be used for cooking, heating, or electricity generation.

Biofuels: Liquid fuels derived from biomass, used primarily for transportation. The two main types of biofuels are:

- **Ethanol**: Made by fermenting sugars from crops like corn, sugarcane, and barley.
- **Biodiesel**: Produced by converting oils from plants (like soybean or canola) or animal fats into a fuel that can replace petroleum-based diesel.

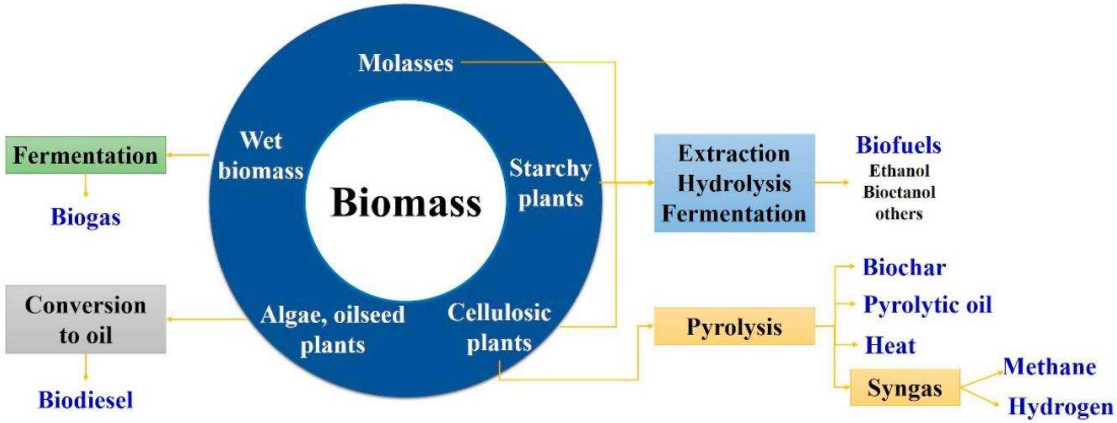
Biomass for Electricity and Heat: Biomass can be burned directly to produce heat or generate electricity through steam turbines. This is commonly used in power plants and industrial facilities.

Biohydrogen: Hydrogen gas produced from organic materials can also be considered a form of bioenergy. It is being explored for its potential use in fuel cells and as a clean energy source.

4. Production Processes

Several processes can be used to convert biomass into bioenergy:

•**Combustion**: Direct burning of biomass to release heat, which can then be used for power generation or heating.



•**Anaerobic Digestion**: The breakdown of organic matter in the absence of oxygen to produce biogas (methane). This is commonly used for waste treatment and manure

•**Fermentation**: The conversion of sugars (from crops like corn or sugarcane) into ethanol by microorganisms.

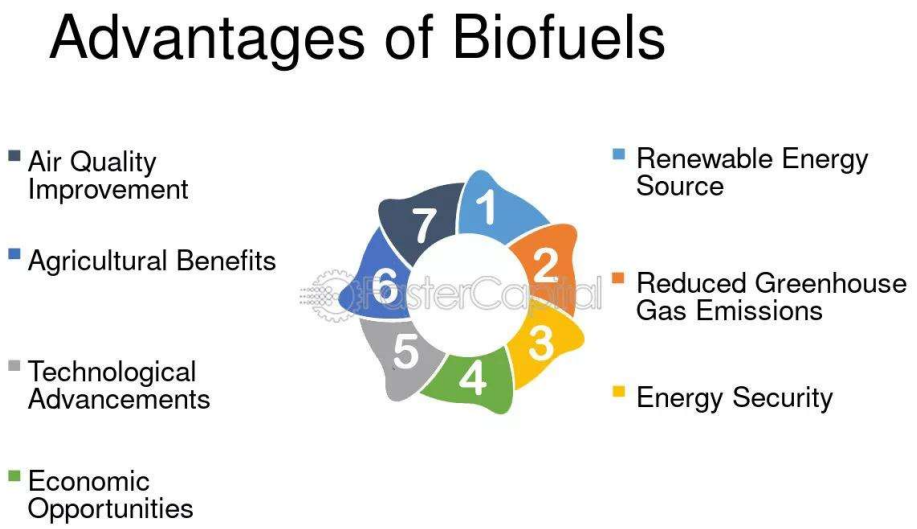
•**Transesterification**: A process used to convert vegetable oils or animal fats into biodiesel.

•**Gasification**: Biomass is heated in a low-oxygen environment to produce syngas (a mixture of hydrogen, carbon monoxide, and methane), which can be used for electricity generation.

•**Pyrolysis**: Biomass is heated in the absence of oxygen, breaking it down into bio-oil, charcoal, and syngas

5. Advantages of Bioenergy

- **Renewable**: Unlike fossil fuels, bioenergy is considered renewable, as long as biomass is replenished (e.g., through sustainable farming or forest management).
- **Carbon Neutral**: Bioenergy is often considered carbon-neutral because the CO_2 released during its combustion is roughly equivalent to the CO_2 absorbed by the plants during their growth.
- **Waste Reduction**: Biomass sources like agricultural and industrial waste can be repurposed into energy, reducing landfill use.



- **Energy Security**: Bioenergy can help reduce dependence on imported fossil fuels and improve national energy security.
- **Economic Benefits**: It can support local economies by creating jobs in agriculture, waste management, and bioenergy industries.

6. Challenges and Limitations

Land Use and Food Security: The use of crops for biofuels (e.g., corn for ethanol) may compete with food production, potentially leading to higher food prices or food scarcity.

Deforestation: Unsustainable harvesting of forest biomass can lead to deforestation and loss of biodiversity.

Energy Efficiency: Some bioenergy production methods (e.g., corn ethanol) have a low net energy gain, meaning the energy used in production is not much lower than the energy produced.

Carbon Emissions: While bioenergy is considered carbon-neutral, the full lifecycle of bioenergy production (from planting crops to processing biomass) can lead to carbon emissions.

Water Use: Bioenergy crops often require significant amounts of water, which can be problematic in water-scarce regions.

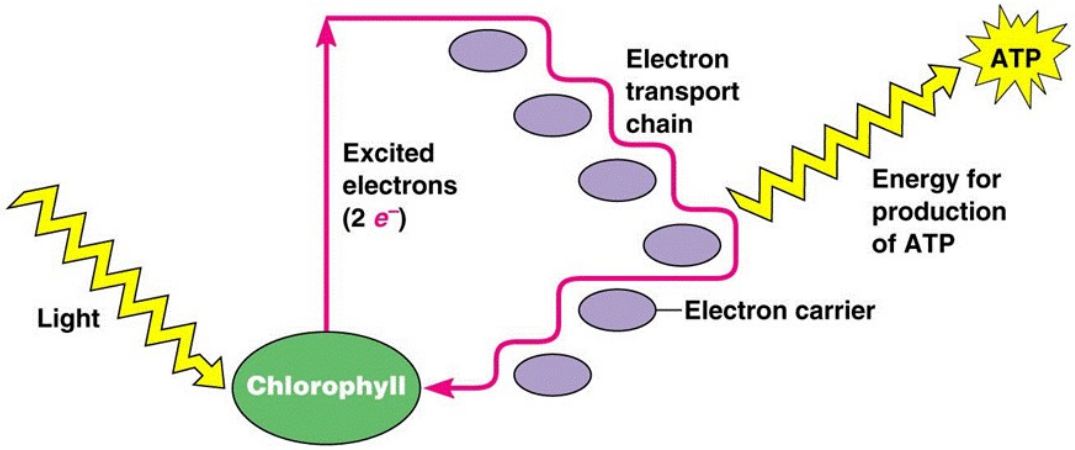
Cost: The production of biofuels and biogas can be expensive compared to traditional fossil fuels, and it often requires government subsidies or incentives to remain competitive.

Mechanism of Photosynthesis

Photosynthesis is the process by which green plants, algae, and certain bacteria convert light energy into chemical energy, storing it in the form of glucose. It primarily takes place in the **chloroplasts** of plant cells, where the pigment **chlorophyll** absorbs sunlight. Here's a breakdown of the key steps involved in photosynthesis:

Light-Dependent Reactions

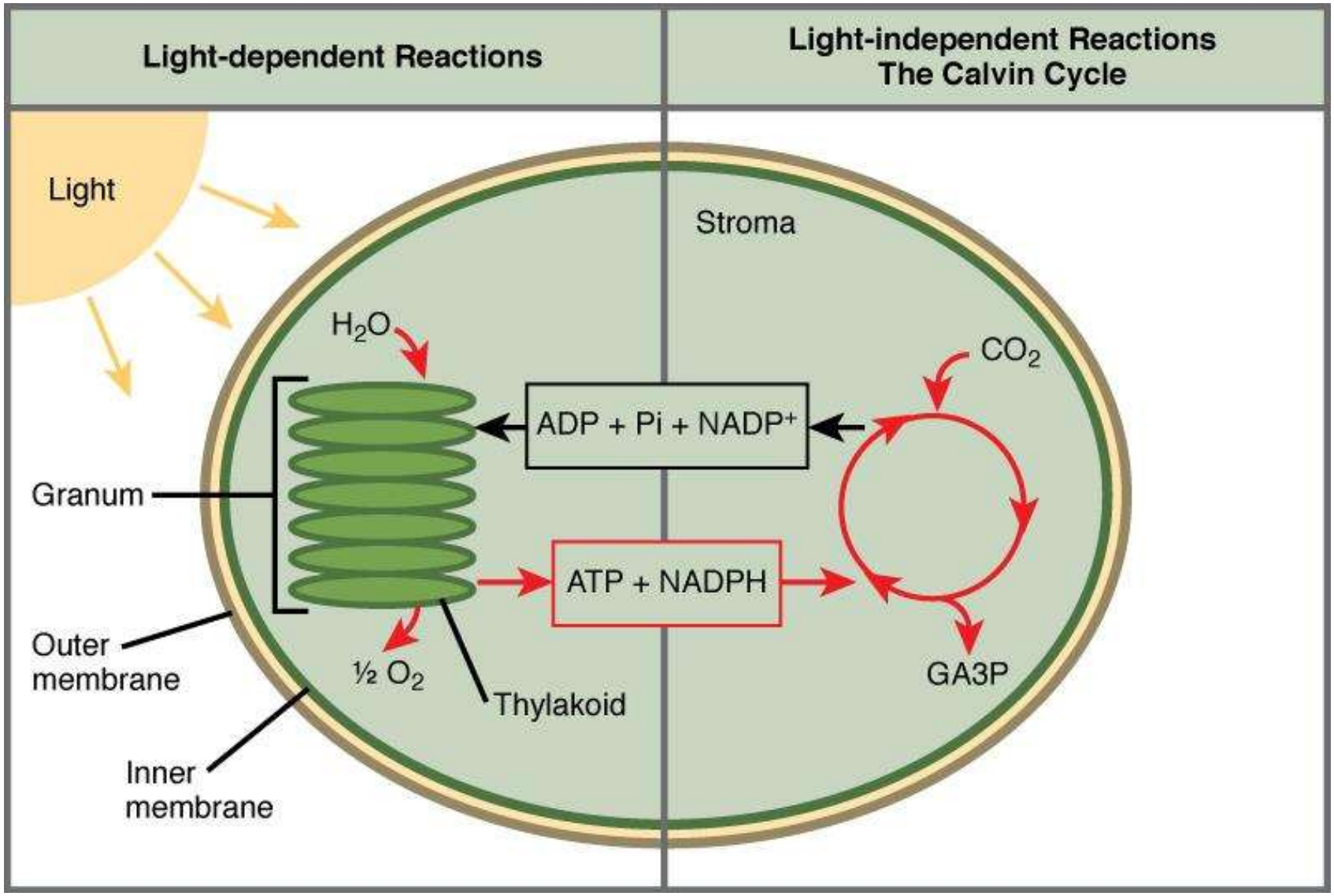
1. Light absorption: Light is absorbed by pigments (chlorophyll, carotenoids) in the thylakoid membrane.
2. Excitation of electrons: Energy from light excites electrons, which are transferred to electron acceptors.
3. Transfer of electrons: Electrons flow through a series of electron carriers in the thylakoid membrane, ultimately resulting in the formation of a proton gradient.



4. ATP synthesis: The proton gradient is used to produce ATP through the process of chemiosmosis.

General Photosynthesis Equation

$$6CO_2 + 6H_2O + \textit{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$$


Light-dependent Reactions
Light-independent Reactions
The Calvin Cycle
Light
Stroma
H_2O
CO_2
ADP + Pi + $NADP^+$
Granum
ATP + NADPH
Outer membrane
½ O_2
Thylakoid
GA3P
Inner membrane

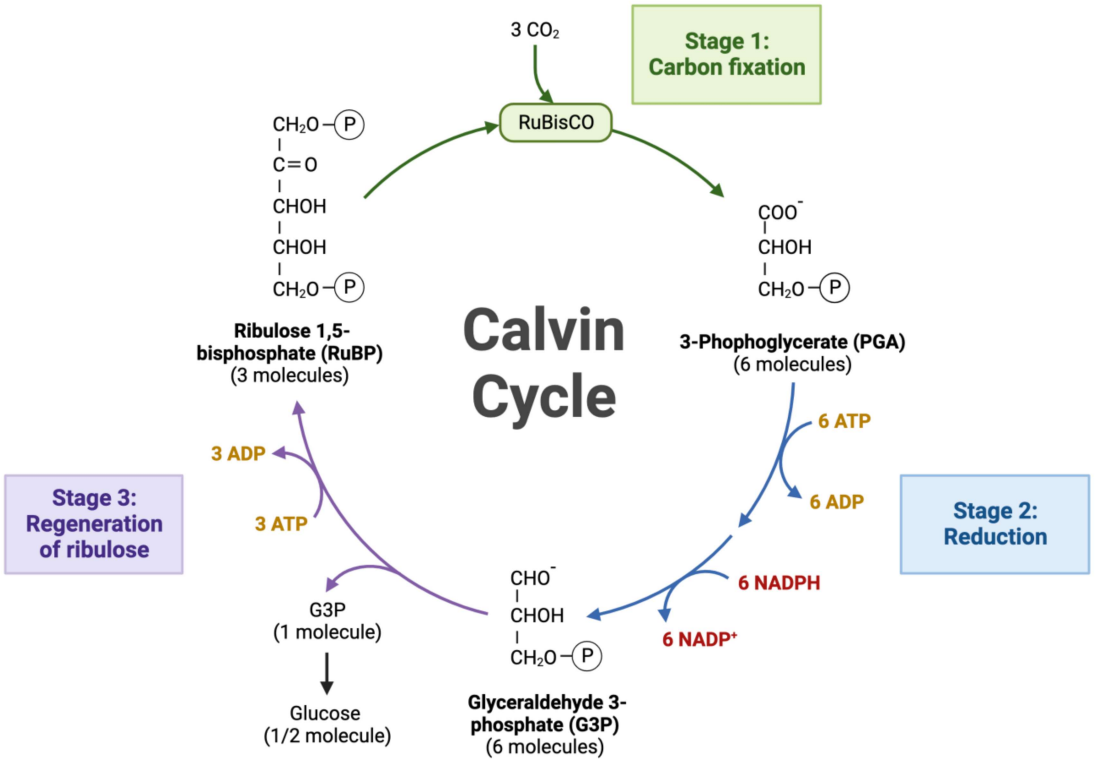
Light-Independent Reactions (Calvin Cycle)

1. Carbon fixation: CO2 is fixed into a 3-carbon molecule (3-phosphoglycerate) via the enzyme RuBisCO.

2. Reduction of 3-phosphoglycerate: 3-phosphoglycerate is reduced to form glyceraldehyde 3-phosphate (G3P) using energy from ATP and NADPH.

3. Regeneration of RuBP: G3P is used to regenerate RuBP, the 5-carbon molecule that binds to CO2.

4. Production of glucose: G3P molecules are linked together to form glucose.



Importance of Photosynthesis

1. Primary production: Photosynthesis produces organic compounds that support life on Earth.

2. Oxygen production: Photosynthesis releases oxygen into the atmosphere.3. Supports food chains: Photosynthesis is the basis of most food chains.

Biofertilizers

Biofertilizers are natural substances that enhance the fertility of the soil and promote plant growth by increasing the availability of nutrients. They contain living microorganisms, such as bacteria, fungi, or algae, that fix nutrients in the soil and make them available to plants. Here's a quick overview of biofertilizers and their components:

Types of Biofertilizers:

1. **Nitrogen Fixing Biofertilizers:**

- **Rhizobium**: Symbiotic bacteria that fix nitrogen in leguminous plants like peas, beans, and lentils.
- **Azotobacter & Azospirillum**: Free-living nitrogen-fixing bacteria that can benefit a wide range of plants.

2.Phosphorus Solubilizing Biofertilizers:PSB (Phosphate Solubilizing Bacteria): Microorganisms that help solubilize phosphate in the soil, making it available for plant uptake.

- **Mycorrhizal Fungi:Arbuscular Mycorrhizae (AMF):** Fungi that form symbiotic relationships with plant roots to improve nutrient uptake, especially phosphorus.
- **Potassium Solubilizing Microorganisms:**Certain bacteria and fungi can help make potassium available to plants by breaking down complex potassium compounds.
- **Plant Growth-Promoting Rhizobacteria (PGPR):**A broad range of bacteria that promote plant growth by producing plant hormones or by suppressing soil pathogens.

Advantages of Biofertilizers:

- **Eco-friendly:** Reduce the need for chemical fertilizers, which can be harmful to the environment.
- **Sustainable Farming:** Improve soil health, support biodiversity, and reduce pollution.
- **Cost-effective:** Lower the costs of farming inputs over time.
- **Soil Health:** Enhance soil structure and increase microbial diversity.

Application Methods:

Seed Treatment: Biofertilizers can be applied directly to seeds before planting.

Soil Application: Biofertilizers are mixed with soil to boost microbial activity.

Root Inoculation: In some cases, they are applied directly to the plant roots to stimulate growth.

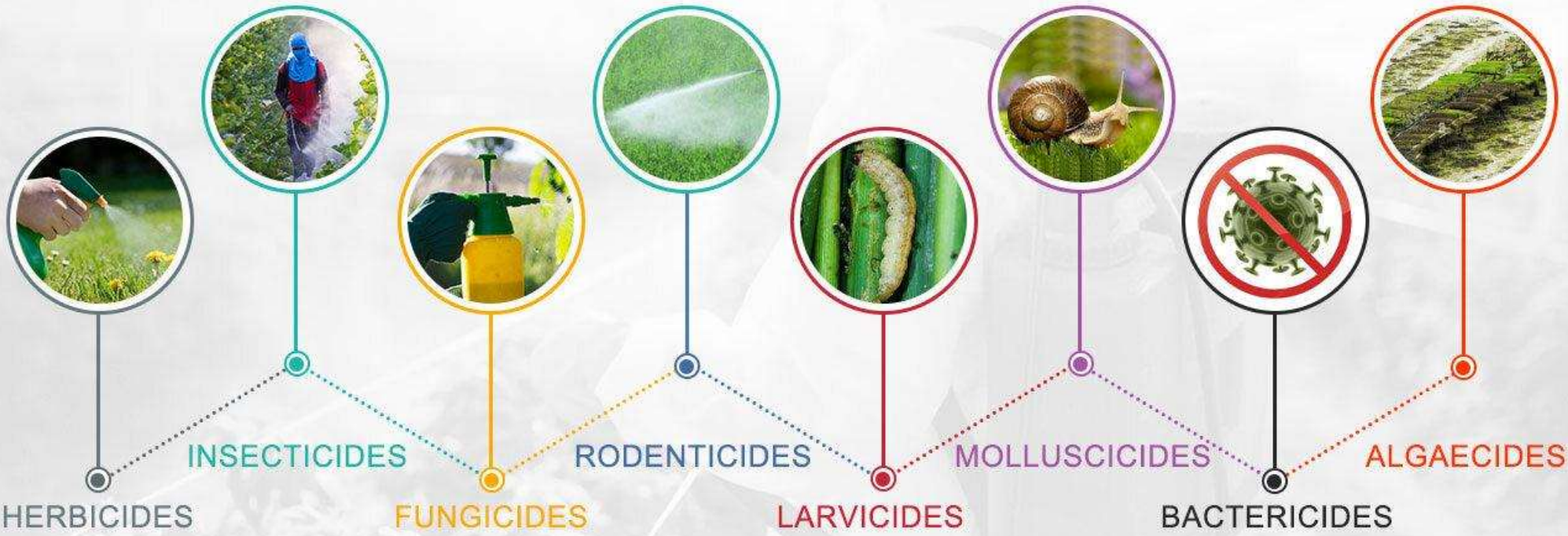
Foliar Spray: Some biofertilizers can be sprayed on leaves to enhance growth and resist pathogens.

Challenges:

Storage & Shelf Life: Biofertilizers need proper storage conditions and have a limited shelf life due to the living microorganisms.

Environmental Conditions: The efficacy of biofertilizers depends on environmental factors like soil temperature, pH, and moisture content.

TYPES OF PESTICIDES USED IN AGRICULTURE



Pesticides are chemicals or biological agents used to control pests, including insects, weeds, fungi, and other organisms that can damage crops, livestock, or structures.
© Raminder Pal Singh/epa/picture alliance

Pesticides

Pesticides are chemicals or biological agents used to control pests, including insects, weeds, fungi, and other organisms that can damage crops, livestock, or structures. Here's a summary of important points regarding pesticides:

1. Types of Pesticides:

•Insecticides:

- **Purpose:** Control insect pests that damage crops, plants, or structures.
- **Examples:**
  - **Organophosphates** (e.g., chlorpyrifos)
  - **Pyrethroids** (e.g., permethrin)
  - **Neonicotinoids** (e.g., imidacloprid)

•Herbicides:

- **Purpose:** Control unwanted weeds that compete with crops for nutrients, water, and light.
- **Examples:**
  - **Glyphosate** (e.g., Roundup)
  - **2,4-D** (a selective herbicide)
  - **Atrazine** (used for corn and other crops)

•Fungicides:

- **Purpose:** Control fungal diseases that can affect crops, plants, and other plants.
- **Examples:**
  - **Sulfur-based fungicides** (e.g., Thiophanate-methyl)
  - **Copper-based fungicides** (e.g., Copper sulfate)
  - **Systemic fungicides** (e.g., Trifloxystrobin)

•**Rodenticides:**

•**Purpose:** Used to control rodent populations, which can spread diseases and cause crop damage.

•**Examples:**

- •**Warfarin**
- •**Bromadiolone**

•**Bactericides:**

•**Purpose:** Control harmful bacteria that affect plants, animals, or the environment.

•**Examples:**

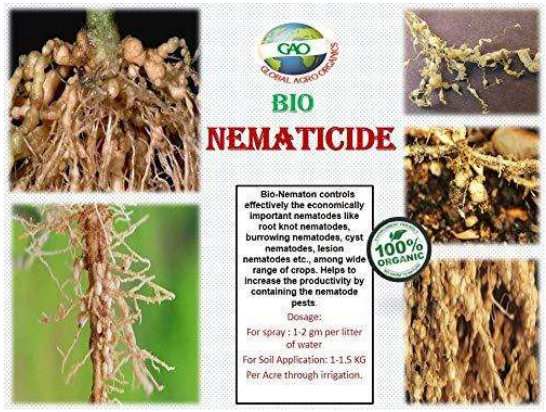
- •**Copper compounds**
- •**Streptomycin** (for plant bacterial diseases)

•**Nematicides:**

•**Purpose:** Control nematodes (microscopic worms) that damage plant roots.

•**Examples**

- •**Methyl bromide**
- •**Carbofuran**



Advantages of Pesticides:

•**Increased Crop Yield:** By controlling pests, they help ensure healthier crops and reduce damage.
•**Protection Against Disease:** Some pesticides protect plants and animals from disease-causing organisms.
•**Economic Benefits:** Higher yields and reduced crop loss contribute to better profits for farmers.
•**Faster Action:** Pesticides can act quickly to control pests, which can be critical for pest outbreaks.

4. Disadvantages and Risks:

•**Environmental Impact:**

- **Pollution:** Runoff from pesticide-treated fields can contaminate soil, water, and air, affecting ecosystems.
- **Non-target Effects:** Pesticides can harm beneficial organisms like bees, birds, and aquatic life.
- **Resistance:** Overuse or misuse of pesticides can lead to pests developing resistance, making the pesticide less effective over time.

•**Human Health Risks:**

- **Toxicity:** Direct exposure to pesticides (through inhalation, ingestion, or skin contact) can cause poisoning, especially for farm workers.
- **Long-term Effects:** Prolonged exposure to certain pesticides may be linked to cancer, reproductive issues, and other health problems.
- **Residue on Food:** Consumers may ingest pesticide residues present on crops if they aren't washed or processed correctly.

•**Biodiversity Loss:** Overreliance on pesticides can lead to a reduction in biodiversity by eliminating both harmful and beneficial organisms.

Safe Use of Pesticides:

•**Read Labels:** Always follow the instructions and safety precautions listed on pesticide labels.

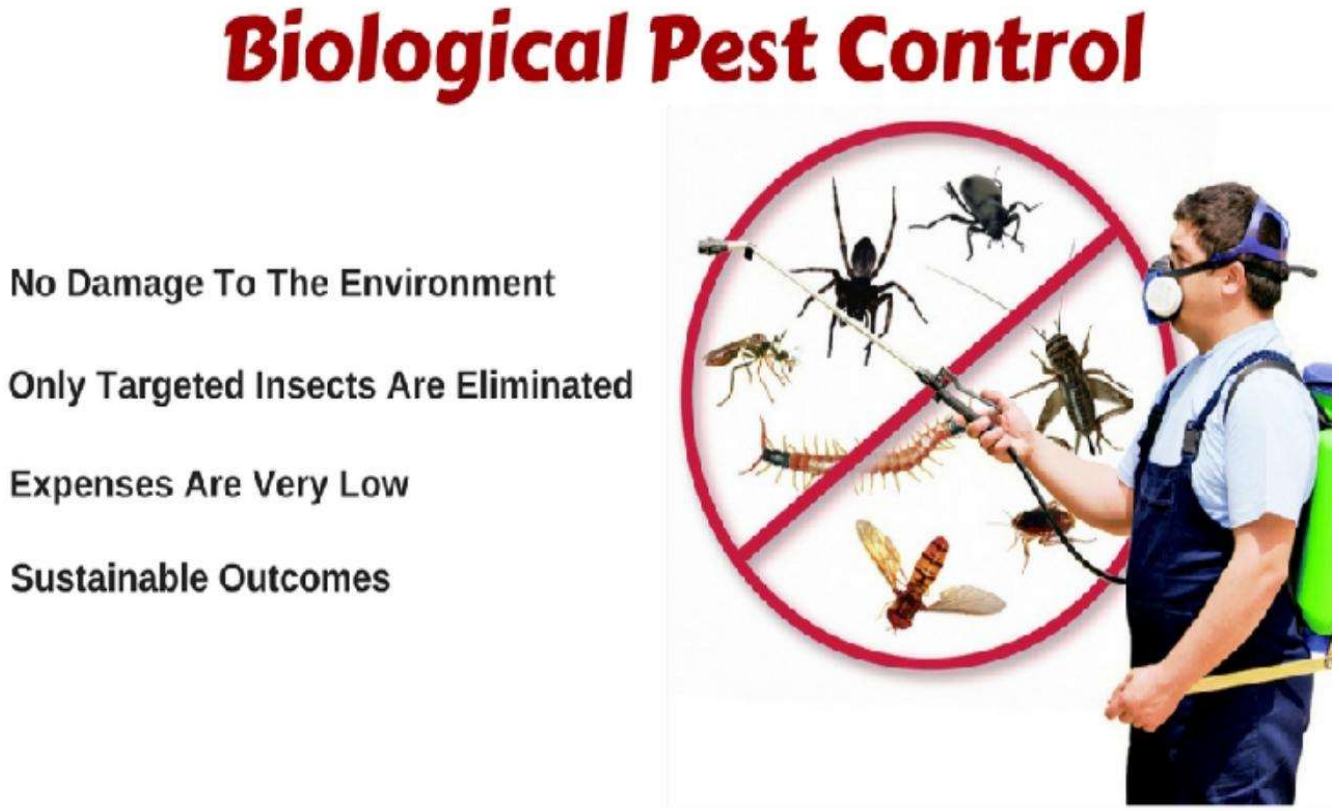
•**Proper Storage:** Store pesticides in a safe, dry, and well-ventilated area away from children and pets.

•**Personal Protective Equipment (PPE):** Use gloves, masks, and protective clothing to minimize exposure.

•**Timing and Dosage:** Apply pesticides at the correct time and in the recommended amounts to avoid overuse.

Biological pest control

Biological pest control is an environmentally friendly and sustainable approach to managing pests by using natural predators, parasites, or pathogens to control pest populations. Instead of relying on synthetic chemicals (like pesticides), biological pest control works by exploiting the natural relationships between organisms, such as predators, prey, and symbiotic relationships. Here's an overview of biological pest control:



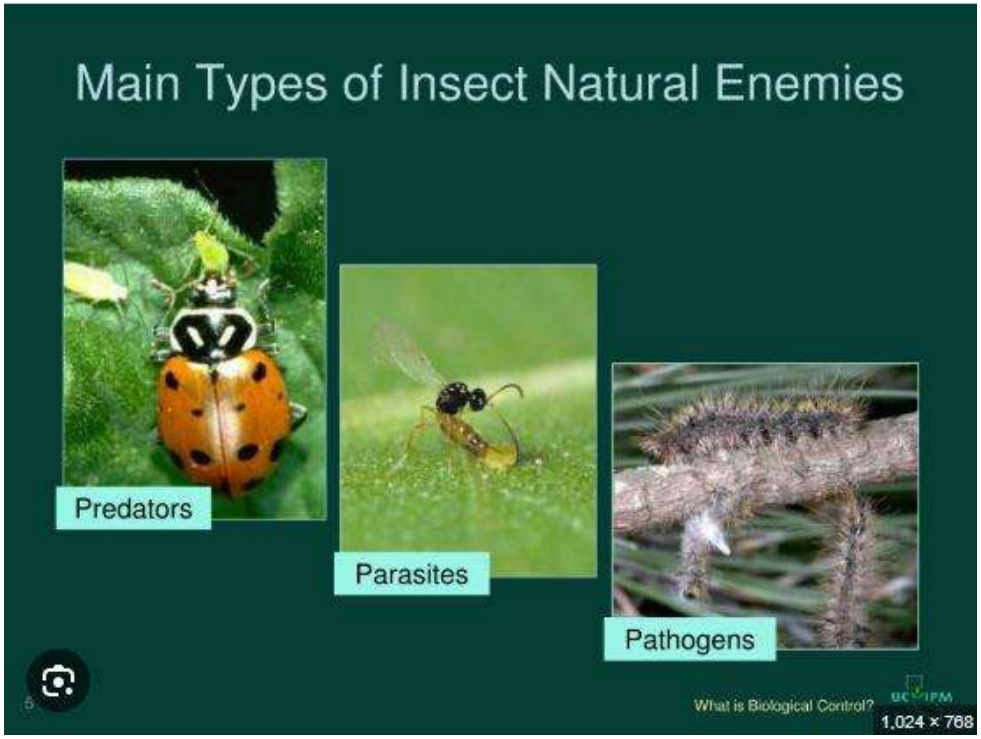
1. Types of Biological Pest Control:

•**Predators:** These are organisms that hunt and kill pests for food.

- **Examples:**
  - **Ladybugs** (ladybird beetles): Prey on aphids and other small insects.
  - **Lacewing larvae:** Consume aphids, mealybugs, and other pests.
  - **Praying mantises:** Eat a wide variety of pests, including insects and small animals.

•**Parasites:** Parasites live in or on a host organism, ultimately killing it to complete their life cycle.

- **Examples:**
  - **Parasitic wasps:** Lay their eggs on or inside pest insects (like caterpillars or aphids). When the eggs hatch, the larvae feed on the host insect, eventually killing it.
  - **Nematodes:** Microscopic roundworms that parasitize and kill soil-dwelling pests like root-feeding nematodes and grubs.



Pathogens: These are microorganisms (bacteria, viruses, fungi) that infect and kill pests.

•Examples:

- **Bacillus thuringiensis (Bt):** A bacterium that produces toxins toxic to specific insects, such as caterpillars (e.g., caterpillars of moths and butterflies).
- **Entomopathogenic fungi:** Fungi like *Beauveria bassiana* infect and kill insects such as termites, whiteflies, and beetles.

Common Examples of Biological Pest Control:

Predators:

Ladybugs: Effective against aphids, scale insects, and other soft-bodied pests.

Lacewings: Both adult and larvae feed on aphids, thrips, and other pests.

Predatory mites: Used to control spider mites, a common pest in agriculture.

Parasites:

Trichogramma wasps: Parasitize and kill the eggs of pest insects like moths and flies.

Tachinid flies: Parasite larvae that attack caterpillars and other insect pests.

Pathogens:

Bacillus thuringiensis (Bt): Bacteria that target caterpillars and larvae of moths and butterflies.

Beauveria bassiana: A fungus that infects and kills a variety of insect pests.

Nematodes:

Steinernema carpocapsae: A nematode that targets soil-dwelling pests like root weevils and grubs.

Target

Applications of Biological Control:

•**Agriculture:** Used to control pests on crops such as vegetables, fruits, and field crops.

•**Horticulture & Gardening:** Home gardeners use natural predators (e.g., ladybugs) to control pests in gardens and on houseplants.

•**Greenhouses:** Biological pest control agents are often employed in enclosed spaces to control pests like whiteflies and aphids.

•**Forestry:** Biological control is used to manage pests that threaten forests, such as the pine beetle or certain moth species.

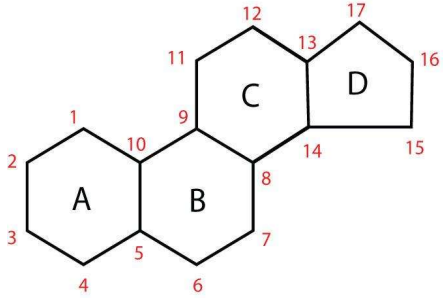
Steroids

Steroids are a broad class of organic compounds that have a common structure consisting of four interconnected carbon rings. They have various functions in the body, ranging from regulating metabolism to acting as signaling molecules. Steroids can be classified into several types, with distinct roles in both humans and animals.

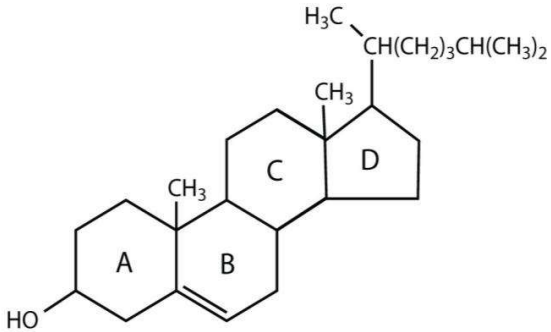
Types of Steroids:

•**Corticosteroids:**

- **Function:** Produced by the adrenal glands, these steroids help manage stress and regulate inflammation, metabolism, and immune responses.
- **Examples:**
  - **Cortisol:** Regulates metabolism, reduces inflammation, and helps the body respond to stress.
  - **Aldosterone:** Helps regulate sodium and potassium levels in the blood, thus maintaining blood pressure.



(a) Steroid skeleton



(b) Cholesterol

•**Anabolic Steroids:**

- **Function:** Synthetic derivatives of testosterone, these steroids are used to promote muscle growth and improve athletic performance. They also play a role in male sexual development.
- **Examples:**
  - **Testosterone:** The primary male sex hormone, crucial for muscle growth, and overall male health.
  - **Synthetic derivatives (e.g., nandrolone, stanozolol):** These mimic the effects of testosterone, helping to increase muscle mass and strength.

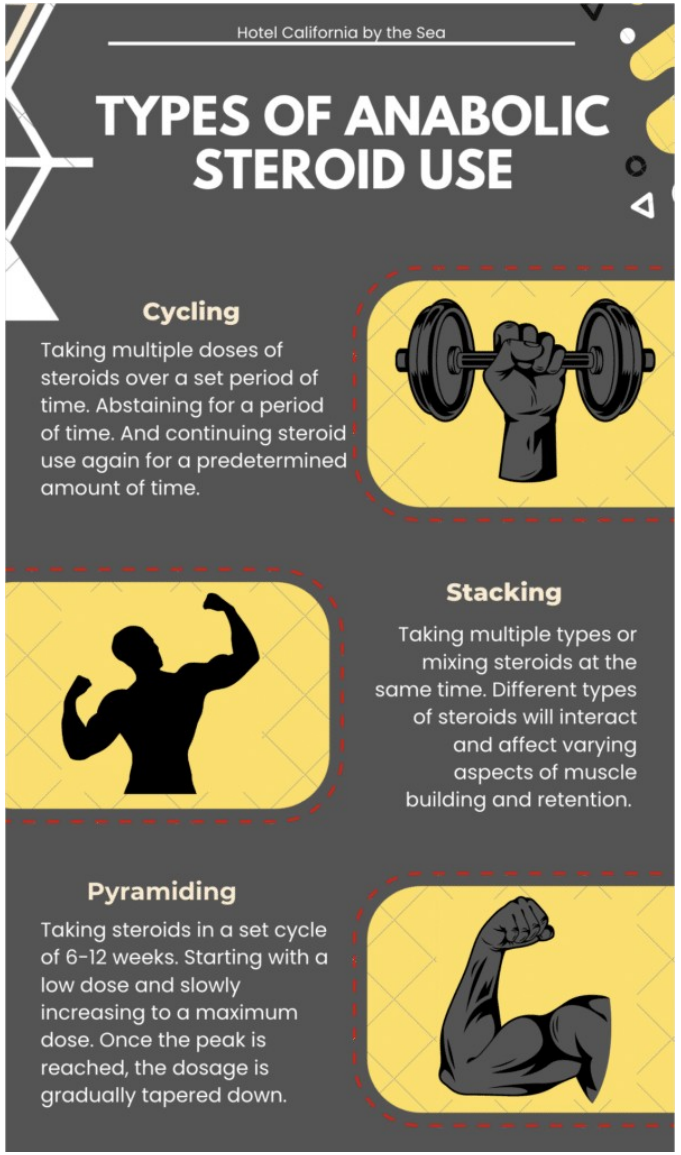
Biological Functions of Steroids:

Metabolic Regulation: Steroids, particularly corticosteroids, regulate metabolic processes, including glucose production and fat storage.

Immune Function: Corticosteroids are important in modulating the immune system and controlling inflammation, which is why they are often used to treat autoimmune diseases and allergies.

Water and Salt Balance: Aldosterone helps control the body's fluid balance, influencing blood pressure and electrolyte levels by regulating sodium and potassium.

Development and Reproduction: Sex hormones like estrogen, progesterone, and testosterone regulate sexual characteristics, fertility, and reproduction.



Uses of Steroids:

•**Medical Uses:**

- **Corticosteroids:** Used to treat inflammatory conditions such as arthritis, asthma, eczema, and autoimmune diseases like lupus. They can also help in treating severe allergies, rashes, or skin conditions.
- **Anabolic Steroids:** Prescribed for patients suffering from muscle-wasting diseases, severe burns, or hormone deficiencies.
- **Sex Hormones:** Used in hormone replacement therapy (HRT) for menopause, transgender hormone therapy, or to manage conditions like endometriosis or polycystic ovary syndrome (PCOS).

•**Performance Enhancement (Abuse of Anabolic Steroids):**

- Some athletes and bodybuilders use anabolic steroids to increase muscle mass, improve strength, and enhance athletic performance. However, this use is illegal in many sports and can lead to serious health issues.

•**Reproductive Health:** Hormonal contraceptives (e.g., birth control pills) often contain synthetic forms of estrogen and progesterone, regulating the menstrual cycle and preventing pregnancy

Vaccines

Vaccines are biological preparations designed to provide immunity to specific diseases. They work by stimulating the immune system to recognize and fight pathogens (such as bacteria, viruses, or parasites) without causing the disease itself. Vaccines have played a pivotal role in public health, preventing countless infections and saving millions of lives. Here's a comprehensive overview of vaccines, including how they work, types, and their importance:

1. How Vaccines Work:

•**Immune System Activation:** Vaccines stimulate the immune system to produce an immune response without causing illness. They typically contain weakened or inactivated pathogens (or parts of them) that the immune system can recognize.

•**Antigen Exposure:** A vaccine contains antigens (components of the pathogen, such as proteins or sugars) that trigger the immune system to produce antibodies. These antibodies help to neutralize or eliminate the pathogen if the body encounters it in the future.

•**Memory Cells:** Once the immune system has responded to the vaccine, it produces memory cells (B-cells and T-cells) that "remember" how to fight that pathogen. If the body is exposed to the pathogen again, the immune system can mount a quicker and more effective response.

HOW VACCINES WORK

Immunize Immunisation Canada
immunize.ca

- Vaccines contain a component or weakened form of a bacteria or virus. The component or weakened form of bacteria or virus in the vaccine does not cause disease.
- Vaccines teach your body's defence (immune) system how to protect you against a specific type of bacteria or virus.
- Different vaccines protect against different types of bacteria or viruses.

- To ensure you have full and lasting protection, you may need to get more than one dose of some vaccines.

Importance of Vaccines:

•**Prevention of Disease:** Vaccines prevent the spread of infectious diseases and reduce the incidence of illness in the population.

•**Herd Immunity:** When a large portion of the population is vaccinated, it helps protect those who cannot be vaccinated (e.g., individuals with weakened immune systems, infants, and the elderly). This is known as herd immunity, and it reduces the overall prevalence of disease.

•**Eradication of Diseases:** Vaccination has led to the complete eradication of some diseases, such as smallpox, and the near-elimination of others, such as polio.

•**Reduced Healthcare Costs:** Vaccines decrease the burden of disease on healthcare systems by preventing hospitalizations and long-term care needs, leading to substantial economic savings.

Challenges in Vaccination

- ❑ Vaccine hesitancy due to misinformation.
- ❑ Limited access in underdeveloped regions.
- ❑ Development of vaccines for rapidly mutating pathogens.

COVID-19

What are the different kinds of vaccine?

RNA

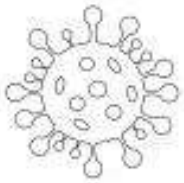
RNA vaccines work by introducing an mRNA sequence (the molecule which tells cells what to build) to the system which is coded for a specific antigen.

DNA

Short for deoxyribonucleic acid, DNA is another of the crucial macromolecules for life. A DNA vaccine involves the direct introduction into appropriate tissues of a plasmid - a doubled-stranded molecule which exists in bacterial cells.

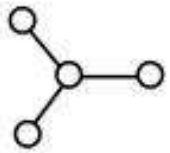
Viral vector

Vaccines use live viruses to carry DNA into human cells.



Virus-like particle

This type of vaccine contains molecules that mimic the virus but are not infectious and, therefore, not a danger. VLP has been an effective way of creating vaccines against diseases such as human papillomavirus (HPV), hepatitis and malaria.



Protein sub-unit

This kind of vaccine uses a part of the virus, in this case the protein component. These vaccines can also be used on almost anyone, including people with weakened immune systems and long-term health problems.

Inactivated virus

These vaccines use the dead version of the virus that causes a disease.

SOURCE: VACCINES.GOV, NEWS MEDICAL LIFE SCIENCES, SCIENCE MEDIA CENTER | NOVEMBER 24, 2020

ALJAZEERA

Common Indian Medicinal Plants and Their Uses

Introduction

India is renowned for its rich biodiversity and traditional knowledge of medicinal plants, which are integral to **Ayurveda**, **Siddha**, and **Unani systems of medicine**. Medicinal plants play a vital role in healthcare and are used for treating various ailments. Below are some common medicinal plants found in India, their active components, and their uses.

1. Tulsi (Holy Basil)

- **Scientific Name:** Ocimum sanctum
- **Active Compounds:** Eugenol, ursolic acid, flavonoids.
- **Uses:**
  - Acts as an adaptogen, improving stress resilience.
  - Antimicrobial and anti-inflammatory properties.
  - Effective in treating cough, cold, bronchitis, and fever.
  - Boosts immunity and has antioxidant effects.

Neem

- **Scientific Name:** Azadirachta indica
- **Active Compounds:** Azadirachtin, nimbin, nimbolide.
- **Uses:**
  - Antiseptic and antifungal agent; used for skin diseases.
  - Helps manage diabetes by lowering blood sugar levels.
  - Used in dental care for oral hygiene.
  - Acts as an insect repellent and supports wound healing.

Amla (Indian Gooseberry)

- **Scientific Name:** Phyllanthus emblica
- **Active Compounds:** Vitamin C, tannins, flavonoids..

Uses:

- Boosts immunity and improves digestion.
- Antioxidant properties help reduce oxidative stress.
- Promotes hair growth and treats scalp issues.
- Beneficial for heart health and managing cholesterol levels.

Ashwagandha (Indian Ginseng)

Scientific Name: Withania somnifera

Active Compounds: Withanolides, alkaloids, saponins.

Uses:

- Reduces stress and anxiety; acts as a natural sedative.
- Enhances strength, stamina, and immunity.
- Supports brain function and memory.
- Used to treat arthritis and promote reproductive health.

Turmeric

- **Scientific Name:** Curcuma longa
- **Active Compounds:** Curcumin, volatile oils.
- **Uses:**
  - Anti-inflammatory and antimicrobial properties.
  - Used in wound healing and skin care.
  - Helps manage arthritis and supports liver health.
  - Boosts immunity and is beneficial in managing respiratory diseases.

6. Ginger

- **Scientific Name:** Zingiber officinale
- **Active Compounds:** Gingerol, shogaol, paradol.
- **Uses:**
  - Aids digestion and relieves nausea and vomiting.
  - Anti-inflammatory properties help in managing arthritis.

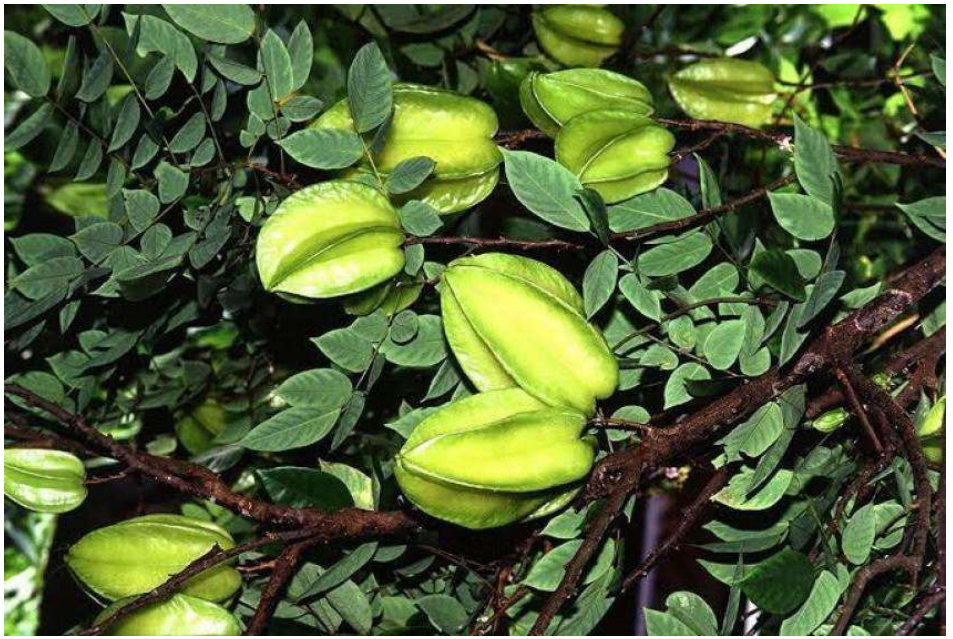
Aloe Vera

- **Scientific Name:** Aloe barbadensis miller
- **Active Compounds:** Aloin, polysaccharides, vitamins.
- **Uses:**
  - Soothes skin irritations and promotes wound healing.
  - Aids in digestion and helps detoxify the body.
  - Acts as a natural laxative.
  - Provides hydration and nourishment to the skin.

Brahmi
Scientific Name: Bacopa monnieri
Active Compounds: Bacosides, alkaloids.
Uses:

- Enhances memory, concentration, and cognitive functions.
- Reduces anxiety and stress.
- Used for managing epilepsy and neurological disorders.
- Acts as an adaptogen, helping the body adapt to stress.

Arjuna



- **Scientific Name:** Terminalia arjuna
- **Active Compounds:** Arjunolic acid, flavonoids, tannins.
- **Uses:**
  - Strengthens the heart muscles and improves cardiac health.
  - Helps manage high blood pressure and cholesterol levels.
  - Acts as an astringent and aids in wound healing.
  - Used in Ayurveda for treating asthma and urinary disorders.

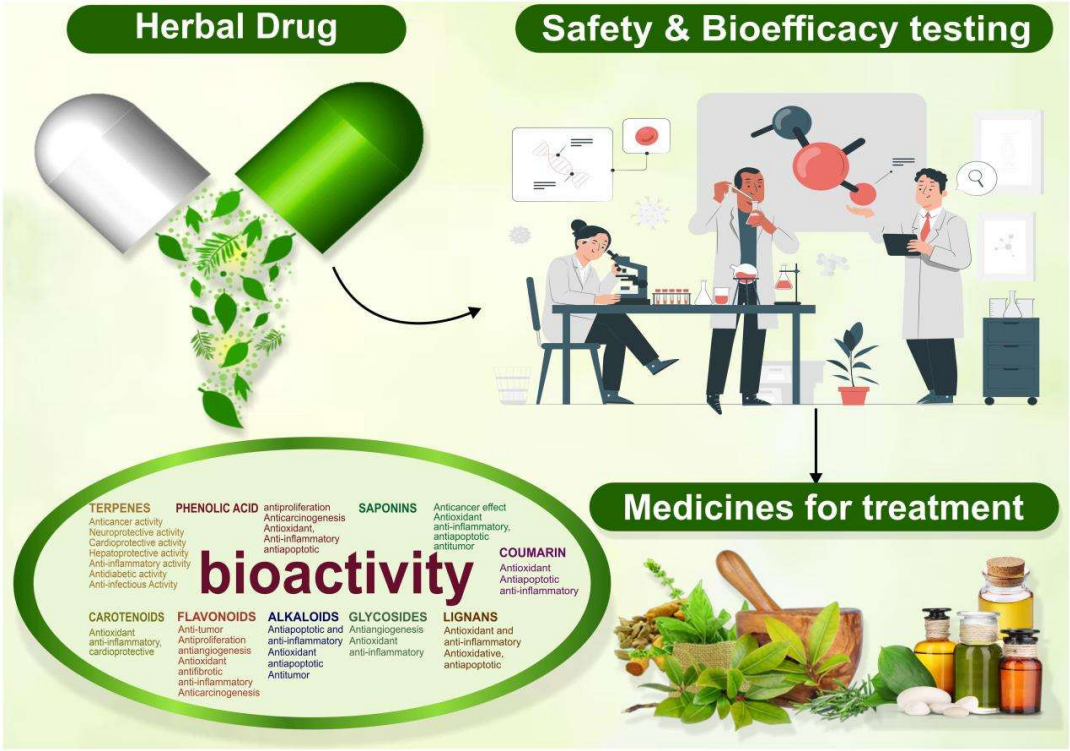
Shatavari

Scientific Name: Asparagus racemosus

Active Compounds: Saponins, alkaloids, flavonoids.

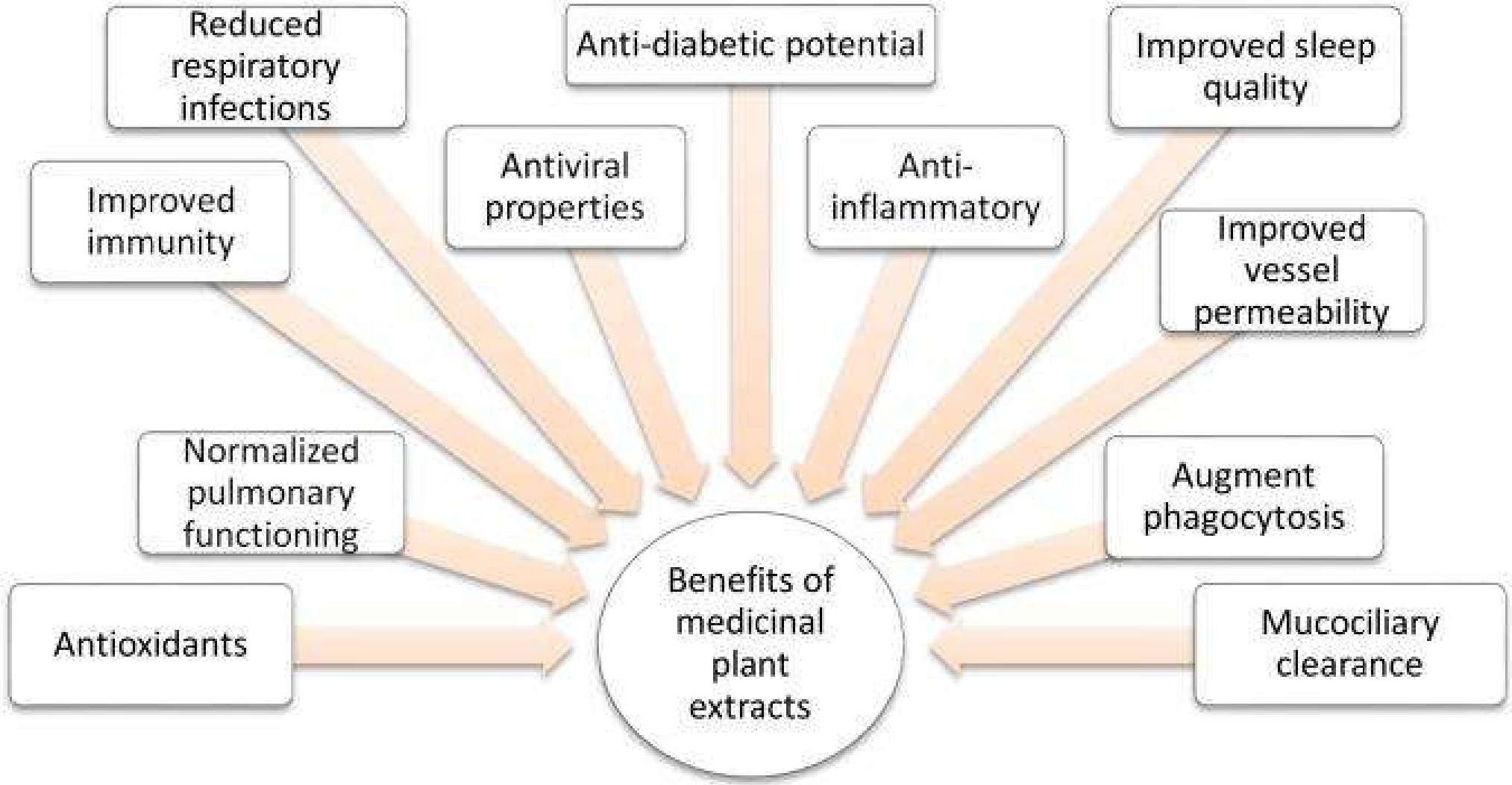
Shatavari

- **Uses:**
  - Promotes reproductive health in women.
  - Acts as a galactagogue, enhancing milk production.
  - Boosts immunity and aids in digestion.
  - Used to manage stress and anxiety.

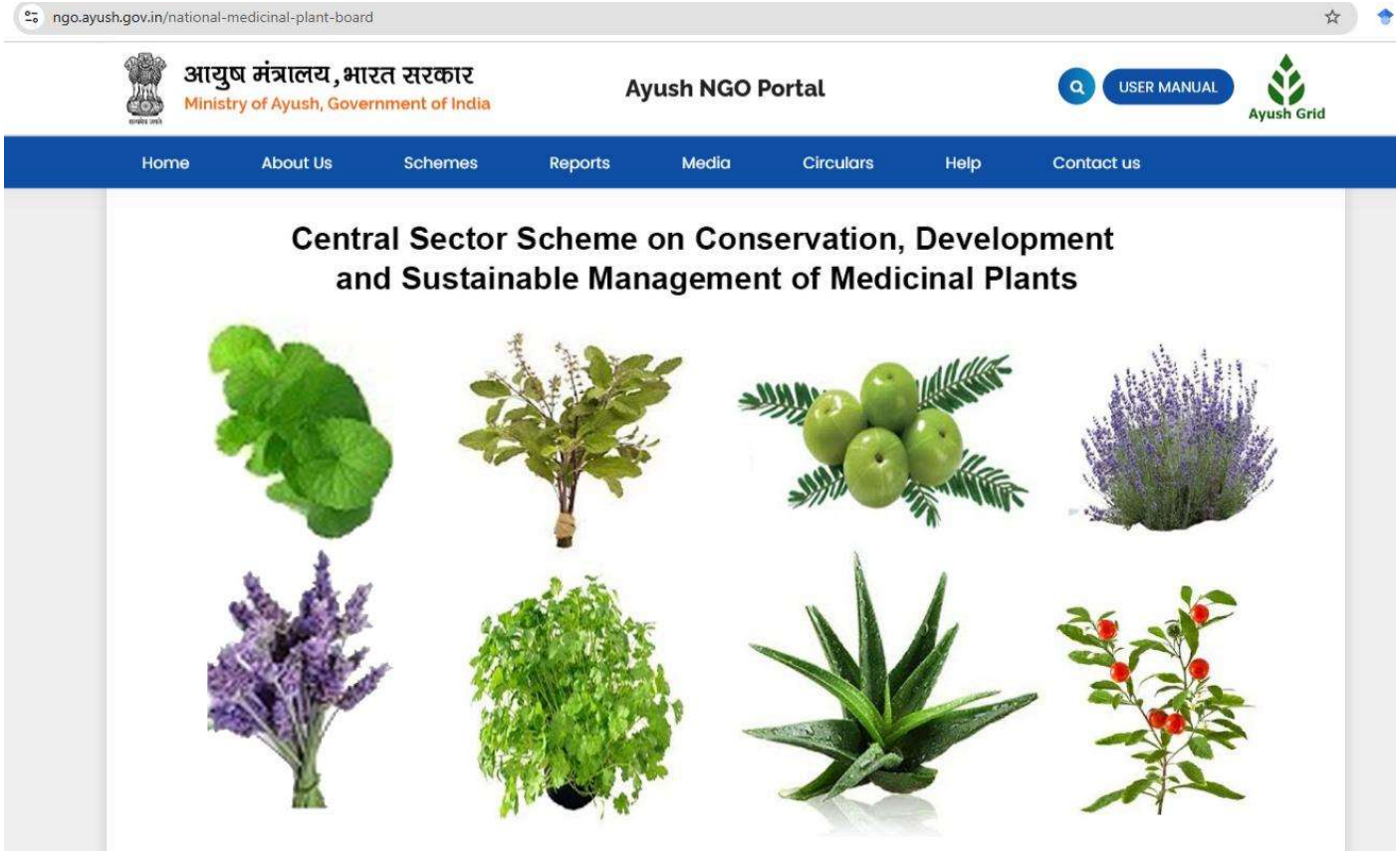


Importance of Medicinal Plants in Modern Healthcare

1. **Source of Pharmaceuticals:**
   - Many modern drugs, such as aspirin (from willow bark) and quinine (from cinchona), are derived from plants.
2. **Holistic Healing:**
   - Used in alternative medicine systems like Ayurveda for treating diseases without side effects.
3. **Economic Importance:**
   - Contribute significantly to India’s herbal industry and export market


Reduced respiratory infections
Anti-diabetic potential
Improved sleep quality
Improved immunity
Antiviral properties
Anti-inflammatory
Improved vessel permeability
Normalized pulmonary functioning
Augment phagocytosis
Antioxidants
Benefits of medicinal plant extracts
Mucociliary clearance

The Ministry of Ayush has established a National Medicinal Plant Board to promote development and cultivation of medicinal plants and to harmonise the development activities between several ministries/departments/organisations. The board has been implementing various schemes and policies for the MSMEs to grow, preserve and market the medicinal plants. Central Sector Scheme for Conservation, Development and Sustainable Management of Medicinal Plants is one such scheme launched by NMPB to focus on cultivation and protection of medicinal plants.



Thank You

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